

LFAENet-TGFS: Text-Guided Frequency Selection for Language-Guided Medical Image Segmentation

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ABSTRACT

Language-guided medical image segmentation (LMIS) pairs a radiological image with its clinical text report and has consistently improved boundary delineation of lesions whose appearance is otherwise ambiguous. Existing LMIS methods, however, fuse vision and text *exclusively in the spatial domain* and ignore the fact that different clinical descriptors live at different spectral scales: “ground-glass opacity” is a low-frequency phenomenon, whereas “sharp nodular rim” is high-frequency. We propose **LFAENet-TGFS**, a Language-guided Frequency-Attention Enhanced Network whose centrepiece is a *Text-Guided Frequency Selection* (TGFS) module. TGFS sits inside every decoder stage of a Haar-DWT-based FAENet backbone and emits per-sub-band soft gates ($\alpha_{LL}, \alpha_{LH}, \alpha_{HL}, \alpha_{HH}$) conditioned on a CXR-BERT text embedding, so the decoder dynamically reweights wavelet components from the clinical prompt itself rather than from a fixed dataset-level prior. An *HH*-drop prior, low-level HF rescaling, language-guided spatial attention and multi-scale deep supervision further sharpen boundaries on small and blurry lesions. We evaluate LFAENet-TGFS across two families of medical imaging under a single unified prompt format: tumor segmentation (brain MRI, breast ultrasound) and pulmonary infection (QaTa-COV19 chest X-ray, MosMedData+ CT). On the tumor datasets, which we re-annotate into this structured prompt format, LFAENet-TGFS attains the best reported accuracy among state-of-the-art methods. On QaTa-COV19 we run a controlled prompt-semantics study in which native prompts substantially outperform empty and shuffled prompts, while MosMedData+ diagnostics show that frequency priors and text conditioning must be tested explicitly rather than assumed universal. Ablations on fusion locus, wavelet sub-bands and MosMedData+ prompt sensitivity corroborate and qualify the design hypotheses behind the method.

1. Introduction

Pixel-level segmentation of pathological regions, such as pulmonary infections, brain tumors and breast lesions, is a fundamental step in computer-aided diagnosis, treatment planning and longitudinal disease monitoring [1, 2]. Despite the progress of CNN- and Transformer-based segmentation networks [3, 4, 5, 6], medical lesions remain difficult to delineate for two recurring reasons. First, lesions vary substantially in shape, size, number and spatial distribution across patients. Second, their boundaries are often low-contrast or visually similar to surrounding anatomy, for example ground-glass opacities blending into normal lung tissue or tumor rims mixing with peritumoral edema. These ambiguities make purely image-driven segmentation unreliable, especially when the target is small, diffuse or weakly contrasted.

In clinical practice, however, medical images are rarely interpreted in isolation. They are accompanied by reports or prompts describing laterality, lesion count, anatomical location and morphology. Recent language-guided medical image segmentation (LMIS) methods exploit this textual prior by fusing report embeddings with visual features. LViT [7] injects language into a vision transformer, while later works use decoder cross-attention [8, 9, 10, 11] or parameter-efficient adapters on foundation models [12, 13]. These methods show that text can guide the model toward the intended target region. Nevertheless, language is usually used as a spatial attention signal: it tells the model *where* to look, but not explicitly *which visual frequency components* should be emphasized to reconstruct the target mask.

Frequency-domain modeling provides a complementary view of medical images. Wavelet decomposition separates a feature map into low-frequency structure and high-frequency boundary details. This is particularly relevant for medical segmentation, where coarse anatomical extent and fine lesion boundaries are both important. Existing frequency-aware models such as XNet [14] and FAENet [15] exploit this property by using wavelet sub-bands to refine

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visual features. More recently, FMISeg [16] combines wavelet-based frequency decomposition with language guidance for LMIS. However, the frequency allocation in these models is still largely determined by the visual architecture or by a fixed low-/high-frequency design, rather than being directly conditioned on the clinical prompt.

Our key observation is that different clinical descriptions imply different frequency needs. A prompt describing “bilateral diffuse ground-glass opacities” mainly requires coarse low-frequency extent, whereas prompts such as “sharp nodular lesion” or “enhancing tumor rim” require stronger directional high-frequency cues. Therefore, text should not only guide spatial attention; it should also modulate the frequency decomposition used for mask reconstruction. This motivates the central question of this work: *can clinical text directly select wavelet sub-bands during segmentation?*

To address this question, we propose **LFAENet-TGFS**, a language-guided frequency-aware segmentation network with *Text-Guided Frequency Selection* (TGFS). Starting from a FAENet-style encoder-decoder backbone, we apply feature-level Haar DWT to decompose intermediate features into four sub-bands: LL, LH, HL and HH. Instead of using a fixed frequency prior, TGFS uses the text embedding to generate sub-band gates ($\alpha_{LL}, \alpha_{LH}, \alpha_{HL}, \alpha_{HH}$) and a token-grounded spatial modulation map. The resulting block allows the clinical prompt to control both *what frequency components* are emphasized and *where* they are applied. The framework is text-encoder agnostic; in our experiments we evaluate both a lightweight trainable text encoder and a CXR-BERT encoder.

We evaluate LFAENet-TGFS on four datasets spanning two families of medical imaging—pulmonary infection (QaTa-COV19 chest X-ray and MosMedData+ CT) and tumor segmentation (brain MRI and breast ultrasound)—under a single unified prompt format. We use QaTa-COV19 as the primary benchmark because it provides clinically meaningful text prompts and allows direct evaluation of whether the model uses prompt semantics. Ablations confirm that native prompts substantially outperform empty and shuffled prompts, demonstrating that the text branch provides genuine semantic guidance. We further study the effect of fusion locus, text encoder choice and frequency-prior design, and include secondary experiments on MosMedData+ to diagnose robustness under CT domain shift. This work makes the following contributions:

- We propose **LFAENet-TGFS**, a language-guided frequency-aware segmentation framework in which a clinical text prompt directly controls the selection of individual Haar wavelet sub-bands inside feature-level frequency refinement blocks, moving language guidance beyond spatial attention toward explicit frequency-domain modulation.
- We introduce the *Text-Guided Frequency Selection* (TGFS) module, which uses a pooled text embedding to generate soft gates over all four wavelet sub-bands and token-level cross-attention to produce a spatial grounding mask, so that the clinical prompt simultaneously governs *which* frequency components are emphasized and *where* they are applied.
- We conduct a systematic study of conditioning locus and frequency-prior handling, showing that encoder-decoder co-conditioning and a learned diagonal-band retention scale both yield consistent improvements over decoder-only and fixed-prior baselines.
- We introduce a unified structured prompt format, re-annotate two tumor datasets into this format, and demonstrate that consistent cross-modal text guidance under a single architecture improves segmentation accuracy and enables controlled prompt-semantics evaluation.
- We evaluate LFAENet-TGFS on four benchmarks spanning pulmonary infection and tumor segmentation; the model achieves state-of-the-art accuracy on the tumor datasets, and prompt-sensitivity ablations confirm that text-guided frequency selection provides genuine, semantically grounded guidance.

2. Related Work

Medical image segmentation has been largely shaped by encoder–decoder architectures that combine semantic abstraction with spatial recovery. U-Net [1] established this template through skip connections between a contracting encoder and an expanding decoder, and UNet++ [3] further strengthened multi-scale feature reuse through nested skip pathways. nnU-Net [2] demonstrated that, with careful configuration of preprocessing, training and inference, this family remains a strong and reliable baseline across medical segmentation tasks. Attention U-Net [17] adds attention gates on the skip connections to suppress irrelevant background responses.

A complementary line of work improves the global reasoning ability of the visual backbone. Building on the Vision Transformer [18], TransUNet [4] combines convolutional features with Transformer token mixing, Swin-Unet [5] introduces hierarchical windowed self-attention, Swin-UNETR [19] extends this design to volumetric medical data, and UCTransNet [6] rethinks U-Net skip connections using channel-wise Transformer interaction. Modernized convolutional backbones such as ConvNeXt [20] remain competitive with these attention-based designs. These methods improve either long-range dependency modeling or cross-scale feature fusion, but they remain image-only systems. As a result, when lesion boundaries are faint, texture is ambiguous, or the target region is only weakly contrasted, the decoder must resolve the segmentation from visual evidence alone. This limitation motivates the use of clinical text as an external semantic prior.

2.1. Language-guided medical image segmentation.

Language-guided medical image segmentation (LMIS) uses textual descriptions, such as lesion type, laterality, anatomical location and extent, to guide visual segmentation. Existing LMIS methods mainly differ in the stage at which text interacts with image features. LViT [7] mixes report and image tokens in a Transformer-based architecture, while RecLMIS [21] uses cross-modal reconstruction to regularize the alignment between visual and linguistic representations. Decoder-oriented methods use language to guide mask reconstruction more directly: LanGuideSeg [8] performs stage-wise language-guided attention, TGCAM [11] calibrates image and report streams with a common attention mechanism, and CausalCLIPSeg [9] introduces causal intervention to reduce spurious text–image correlations.

Recent works also explore more efficient or more deployable forms of multimodal adaptation. MAdapter [10] learns lightweight bidirectional adapters between frozen vision and language encoders, while SGSeg [22] studies text-free inference through generated self-guidance, and STPNet [23] retrieves scale-aware textual descriptions from a medical text repository so that language guides training while inference remains text-free. Such multimodal alignment builds on vision–language pretraining: CLIP [24] aligns image and text embeddings at scale, and medical adaptations such as GLoRIA [25], MedCLIP [26] and PubMedCLIP [27] specialize this alignment to clinical data. Foundation-model-based systems, including LGA [12] and MedCLIP-SAMv2 [13], translate medical language into prompts for large segmentation models such as SAM [28] and its medical variant MedSAM [29].

Although these methods differ in architecture, most of them use text primarily as a semantic or spatial selector. The prompt specifies what should be segmented, or where the model should attend, while the visual feature representation is still fused mainly in the spatial domain. In this work, we ask a more specific question: can clinical text also decide which frequency components should be emphasized when reconstructing the lesion mask?

2.2. Frequency-domain segmentation.

Frequency-domain learning provides a complementary way to represent visual evidence. Instead of processing only spatial feature maps, frequency-aware methods decompose the signal into components associated with coarse structure, texture and boundary detail. DCT- and Fourier-based attention modules, such as FcaNet [30] and FsaNet [31], learn to emphasize informative spectral responses. These approaches are useful for spectral modeling, but their global basis is less explicitly localized around lesion boundaries.

Wavelet-based methods are especially relevant for segmentation because they preserve spatial–frequency locality. Multi-level wavelet CNNs [32] embed the discrete wavelet transform into the network to enlarge the receptive field without discarding detail, and XNet [14] separates low- and high-frequency paths for biomedical image segmentation. More recent designs continue this direction: WaveFormer [33] couples a wavelet-driven feature representation with a Transformer for efficient volumetric segmentation, and SVD-based linear attention in the frequency domain has been shown to improve boundary detail in medical segmentation [34]. Frequency-domain analysis has also been revisited specifically for lesion segmentation, where reformulating brain-tumor segmentation in the spectral domain improves the recovery of fine structures [35]. FAENet [15] further decomposes intermediate features into four Haar sub-bands, LL , LH , HL and HH , and refines them through intra- and inter-sub-band channel attention. This decomposition is well aligned with medical lesion segmentation: low-frequency responses often describe coarse lesion extent, directional high-frequency responses preserve boundary cues, and the diagonal HH component may contain either useful fine detail or noise. Therefore, frequency priors should be treated as design hypotheses rather than fixed rules.

FMISeg [16] is the closest prior work to ours because it combines frequency-domain modeling with language-guided medical segmentation. However, its frequency representation follows a fixed low-/high-frequency design, where directional high-frequency components are grouped before language interaction. By contrast, LFAENet-TGFS keeps the four Haar sub-bands available inside feature-level frequency refinement and lets the clinical prompt modulate them

directly. Pooled text features generate sub-band gates over $LL/LH/HL/HH$, while token-level features produce a spatial grounding mask. Thus, our goal is not to introduce a new wavelet transform or a new language encoder, but to turn frequency selection from a static visual prior into a dynamic, language-conditioned operation.

3. Methodology

3.1. Overview

Given a medical image $I \in \mathbb{R}^{H \times W \times C}$ and a clinical text prompt T , language-guided medical image segmentation aims to predict a binary lesion mask $\hat{Y} \in [0, 1]^{H \times W}$, where H , W , and C denote the image height, width, and channel count. We propose LFAENet-TGFS, a language-conditioned frequency-aware segmentation network whose centrepiece is *Text-Guided Frequency Selection* (TGFS). The key question addressed by our design is not whether frequency modeling is useful by itself, but *where* language should enter frequency reasoning. Instead of using text only as a spatial attention signal, LFAENet-TGFS lets the prompt modulate the four Haar wavelet sub-bands LL , LH , HL , and HH inside feature-level frequency refinement blocks. A single architecture is used across all four datasets in this paper; only dataset-specific training settings (loss weights and data augmentation) differ, and these are reported in the implementation details (Sec. 4.1).

Figure 1 gives the overall architecture. The visual stream is an encoder–decoder built on an ImageNet-pretrained ResNet-50 encoder with FAENet-style feature-level wavelet refinement at each stage. The text stream encodes the clinical prompt with a frozen domain-specific clinical language model and produces two semantic signals: a pooled text vector for sub-band gating and token embeddings for spatial grounding. Text-conditioned TGFS blocks are inserted in the encoder, bottleneck, and decoder, so language conditions early feature extraction, high-level compression, and mask reconstruction together. The four sub-bands $LL/LH/HL/HH$ are preserved inside every TGFS block, and the model never collapses them into a two-branch LF/HF stream.

This design keeps the method aligned with our motivation: the contribution is not a new visual backbone, but language-conditioned control over the frequency reasoning process. It also distinguishes LFAENet-TGFS from raw-image LF/HF dual-branch designs, in which wavelet decomposition is applied to the input image to form separate low- and high-frequency image streams that interact with language in a late-fusion decoder. By contrast, LFAENet-TGFS applies Haar DWT inside intermediate feature-level blocks and lets the prompt directly control wavelet sub-band selection during refinement. We first isolate the inherited visual frequency component (Sec. 3.2), then describe the text-guided selection block that is the core of LFAENet-TGFS (Sec. 3.3), and finally the conditioning configuration (Sec. 3.4) and training objective (Sec. 3.5).

3.2. FAENet-Style Visual Frequency Backbone

We adopt an ImageNet-pretrained ResNet-50 as the visual encoder and a FAENet-style frequency-attention decoder as the inherited visual base; both are used without architectural modification. Its frequency attention module, denoted FreqA, models spectral–spatial interactions through Haar wavelet decomposition, intra-sub-band attention, cross-sub-band attention, and inverse reconstruction. This block is the visual base; the novelty of LFAENet-TGFS is the language-conditioned sub-band modulation built on top of it.

For an intermediate visual feature map $X \in \mathbb{R}^{C \times H \times W}$, Haar DWT decomposes X into four sub-bands:

$$X_{LL}, X_{LH}, X_{HL}, X_{HH} = \text{DWT}(X), \quad (1)$$

where LL captures coarse low-frequency structure, LH and HL encode directional high-frequency responses, and HH contains diagonal high-frequency detail. Each sub-band has spatial resolution $H/2 \times W/2$.

The FreqA block first applies inner-component channel attention (ICCA) within each sub-band:

$$\bar{X}_k = \text{ICCA}_k(X_k), \quad k \in \{LL, LH, HL, HH\}, \quad (2)$$

where k indexes the four sub-bands. It then applies cross-component channel attention (CCCA) to exchange information across frequency components:

$$[\hat{X}_{LL}, \hat{X}_{LH}, \hat{X}_{HL}, \hat{X}_{HH}] = \text{CCCA}([\bar{X}_{LL}, \bar{X}_{LH}, \bar{X}_{HL}, \bar{X}_{HH}]), \quad (3)$$

and inverse DWT reconstructs the spatial feature with a residual connection:

$$Y = \text{iDWT}(\hat{X}_{LL}, \hat{X}_{LH}, \hat{X}_{HL}, \hat{X}_{HH}) + X, \quad (4)$$

where Y is the refined feature map and X is added back as a residual. This FreqA pathway is purely visual and does not use text. Our method keeps this spectral–spatial reasoning principle, but replaces visual frequency refinement at the conditioned stages with TGFS, where the prompt directly modulates the sub-bands before cross-band mixing.

3.3. Text-Guided Frequency Selection

Prompt encoding. We encode the clinical prompt $T = \{t_1, \dots, t_N\}$ with a frozen domain-specific clinical language model into token embeddings $E_T \in \mathbb{R}^{N \times d}$ and a pooled embedding $e_T \in \mathbb{R}^d$:

$$E_T, e_T = \text{TextEnc}(T), \quad (5)$$

where N is the number of tokens and d is the text embedding dimension. The framework is text-encoder agnostic; we use a frozen clinical language encoder because clinical prompts are domain-specific and a pretrained clinical encoder provides a strong semantic prior without additional training cost. The two text outputs play complementary roles, as summarized in Figure 2: the pooled embedding e_T provides global semantic conditioning and predicts sub-band gates over $LL/LH/HL/HH$, while the token embeddings E_T preserve word-level information and form a token-grounded spatial mask M_s . This separation follows the motivation from the Introduction: clinical text should guide both *what* visual frequency components are emphasized and *where* the response should be applied.

Sub-band gating. The TGFS block is the core module of LFAENet-TGFS and is detailed in Figure 3. Given a visual feature X and text representations (e_T, E_T) , TGFS first applies local pre-refinement and Haar decomposition:

$$X' = \phi_{\text{local}}(X), \quad X_{LL}, X_{LH}, X_{HL}, X_{HH} = \text{DWT}(X'), \quad (6)$$

where $\phi_{\text{local}}(\cdot)$ denotes a lightweight convolutional block. After ICCA, TGFS predicts channel-wise gates for the four sub-bands. Let $g(X')$ denote a global visual descriptor obtained by spatial pooling. The gates are computed as

$$[\alpha_{LL}, \alpha_{LH}, \alpha_{HL}, \alpha_{HH}] = \sigma(\text{MLP}([g(X'), e_T])), \quad (7)$$

where σ is a sigmoid, MLP is a small multilayer perceptron, $[\cdot, \cdot]$ denotes concatenation, and $\alpha_k \in [0, 1]^C$ is the gate for sub-band k . Each sub-band is then modulated independently:

$$\tilde{X}_k = \alpha_k \odot \text{ICCA}_k(X_k), \quad k \in \{LL, LH, HL, HH\}, \quad (8)$$

where $\odot$ denotes channel-wise (Hadamard) multiplication. This is the central operation of LFAENet-TGFS. Unlike text fusion that concatenates text with a spatial feature map after visual processing, Eq. (8) lets the prompt influence the relative contribution of coarse structure and high-frequency detail *before* cross-band interaction. The gated sub-bands are passed through CCCA:

$$[Z_{LL}, Z_{LH}, Z_{HL}, Z_{HH}] = \text{CCCA}([\tilde{X}_{LL}, \tilde{X}_{LH}, \tilde{X}_{HL}, \tilde{X}_{HH}]). \quad (9)$$

The four-band representation is preserved throughout; cross-band interaction only allows the sub-bands to exchange information before reconstruction.

Token-grounded spatial modulation. The frequency gates in Eq. (7) provide global semantic selection over sub-bands. To complement them with spatial grounding, TGFS uses token embeddings to generate a mask over the wavelet-refined feature. Let Z denote the CCCA-refined representation. We project Z into visual queries Q_v and project token embeddings E_T into text keys and values (K_t, V_t) :

$$A = \text{softmax}\left(\frac{Q_v K_t^\top}{\sqrt{d}}\right), \quad R = A V_t, \quad (10)$$

where A is the cross-attention map, R is the attended text response, and d is the projection dimension. The response R is converted into a token-grounded spatial mask:

$$M_s = \sigma(\psi([Z, R])), \quad M_s \in [0, 1]^{1 \times H/2 \times W/2}, \quad (11)$$

where $\psi(\cdot)$ is a lightweight projection head. The mask modulates the CCCA-refined feature:

$$Z_s = M_s \odot Z. \quad (12)$$

The result is split back into four sub-bands and reconstructed with inverse DWT:

$$Y = \text{iDWT}(Z_{LL}^s, Z_{LH}^s, Z_{HL}^s, Z_{HH}^s) + X. \quad (13)$$

Thus, TGFS uses text through two complementary pathways: pooled text controls frequency selection, while token-level text controls spatial localization.

3.4. Conditioning Locus and Frequency-Prior Handling

Conditioning locus. We condition both the encoder and the decoder with text. On the decoder side, conditioning is applied through the TGFS gates and the token-grounded mask. On the encoder side, text interacts with visual features through cross-attention before frequency refinement. The full placement is

$$S_{\text{both}} = \{\text{Enc}_1, \dots, \text{Enc}_4, \text{Bottleneck}, \text{Dec}_4, \dots, \text{Dec}_1\}, \quad (14)$$

where S_{both} is the set of conditioned stages, so that language conditions early visual frequency extraction, high-level semantic compression, and low-level mask reconstruction. Decoder-only and encoder-only conditioning are studied as ablations (Sec. 4.4).

Frequency-prior handling. The diagonal HH sub-band may carry either useful boundary detail or noise. Rather than imposing a hard prior that the highest-frequency component is always noise, we expose HH handling with three policies:

$$Z_{HH}^{\text{out}} = \begin{cases} 0, & \text{zero,} \\ Z_{HH}, & \text{keep,} \\ \lambda_{HH} Z_{HH}, & \text{learned,} \end{cases} \quad (15)$$

where λ_{HH} is a learned scale. The main model uses the learned policy, which lets the network decide per stage how much diagonal high-frequency response to retain; zero and keep are reported as ablations.

3.5. Training Objective

The model is trained with a pixel-wise segmentation loss combining binary cross-entropy, Dice, and a boundary term [36]:

$$\mathcal{L}_{\text{seg}} = \lambda_{\text{BCE}} \mathcal{L}_{\text{BCE}} + \lambda_{\text{Dice}} \mathcal{L}_{\text{Dice}} + \lambda_{\text{bd}} \mathcal{L}_{\text{bd}}, \quad (16)$$

where $\mathcal{L}_{\text{BCE}}$, $\mathcal{L}_{\text{Dice}}$, and $\mathcal{L}_{\text{bd}}$ are the binary cross-entropy, Dice, and boundary losses, and λ_{BCE} , λ_{Dice} , λ_{bd} are their weights. Deep supervision [37] adds the same segmentation loss at auxiliary decoder outputs, and a grounding term supervises the token-grounded mask M_s with the downsampled ground truth:

$$\mathcal{L} = \mathcal{L}_{\text{seg}}(\hat{Y}, Y) + \sum_s w_s \mathcal{L}_{\text{seg}}(\hat{Y}_s, Y) + \lambda_g \mathcal{L}_g(M_s, Y), \quad (17)$$

where Y is the ground-truth mask, $\hat{Y}_s$ are auxiliary predictions at decoder stage s with weight w_s , $\mathcal{L}_g$ is a binary cross-entropy on the grounding mask, and λ_g is its weight. Loss weights and dataset-specific settings are given in Sec. 4.1.

4. Experiments

We organize the experiments around three questions. First, does adding a language-conditioned frequency decoder improve a FAENet-style visual frequency backbone? Second, does the model use the prompt content, rather than merely the presence of text tokens? Third, is the structured prompt format a valid interface for extending language-guided segmentation from pulmonary infection datasets to Brain MRI and Breast Ultrasound? Unless otherwise stated, Dice and IoU are reported in percentage points.

Table 1

Dataset and prompt protocol summary. Brain MRI and Breast Ultrasound use our structured prompt extension, designed to match the QaTa/MosMed style of laterality, lesion count, and coarse image-region location.

Dataset	Modality	Split	Prompt source	Role in experiments
Brain MRI	MRI	1462/1002/600	Ours: structured tumor pseudo-reports	Cross-modality benchmark and prompt-template transfer
Breast US	US	599/50/113	Ours: structured lesion pseudo-reports	Cross-modality benchmark and prompt-template transfer
QaTa-COV19	CXR	5716/1429/2113	Existing structured infection pseudo-reports	Primary prompt-semantics and mechanism benchmark
MosMedData+	CT	2183/273/273	Existing structured infection pseudo-reports	CT-domain robustness and frequency-prior diagnostic

Table 2

Comparison with state-of-the-art methods across four datasets under the main structured-prompt protocol. Best result per column is in **bold**. “–” denotes a result not reported or not yet evaluated under this protocol.

Method	Brain MRI		Breast US		QaTa-COV19		MosMedData+	
	Dice	IoU	Dice	IoU	Dice	IoU	Dice	IoU
UNet	72.16	60.51	71.54	62.34	78.45	68.76	64.58	50.73
nnUNet	77.76	67.71	73.77	63.77	80.42	70.81	72.59	60.36
TransUNet	80.83	71.52	80.60	71.68	78.63	69.13	71.24	58.44
VT-MFLV	84.63	75.37	78.05	67.15	83.34	72.09	75.61	63.98
STPNet	79.66	69.62	71.25	60.13	80.63	71.42	76.18	63.41
MedCLIP-SAMv2	80.03	70.71	78.87	69.08	–	–	–	–
FMISeg	–	–	82.20	73.30	91.21	83.84	79.30	65.71
LFAENet-TGFS (ours)	86.51	78.11	83.86	76.15	90.04	81.88	80.10	66.68

4.1. Datasets, Prompt Protocols, and Evaluation Settings

We evaluate on four binary lesion segmentation datasets spanning pulmonary infection and tumor segmentation. QaTa-COV19 is used as the primary controlled benchmark because its sample-specific structured reports allow direct prompt interventions. MosMedData+ is used as a secondary CT-domain robustness and frequency-prior diagnostic. Brain MRI and Breast Ultrasound are used for cross-modality evaluation and, importantly, for testing prompt-template transfer.

The main benchmark uses a unified structured-prompt protocol. For QaTa-COV19 and MosMedData+, prompts describe laterality, lesion count, and coarse anatomical location. For Brain MRI and Breast Ultrasound, we construct analogous structured pseudo-reports from the available lesion-location information. These prompts are not claimed to be free-form radiology reports; they are a controlled text interface for supervised language-guided segmentation. Sec. 4.3 separately compares this structured format with MedCLIP-SAMv2-style class/descriptive prompts.

For all datasets, checkpoints are selected by validation Dice and the final binarization threshold is chosen on the validation split by sweeping $\{0.35, 0.40, 0.45, 0.50, 0.55\}$. The test set is never used for checkpoint or threshold selection. Dice and IoU are computed per image and then averaged over the test set. Missing values are denoted by “–” and correspond to runs that are planned but not yet completed under the stated protocol.

4.2. Main Four-Dataset Comparison

Table 2 gives the main comparison under the structured prompt protocol. We keep this table compact and place the mechanism ablations in later subsections. MedCLIP-SAMv2 is not mixed into the fair structured-prompt comparison unless it is evaluated with the same test prompts; its original prompt protocol is reported separately in Table 3.

Table 3

Prompt-template transfer on Brain MRI and Breast Ultrasound. The table separates model architecture from prompt format. MedCLIP-SAMv2-style prompts are short class/descriptive prompts used by the CLIP-SAM pipeline, whereas our structured prompts encode lesion count and coarse image-region location.

Model	Learning setting	Test prompt format	Brain MRI		Breast US	
			Dice	IoU	Dice	IoU
MedCLIP-SAMv2	zero/one-shot	original MedCLIP-SAMv2 prompt	80.03	70.71	78.87	69.08
MedCLIP-SAMv2	zero/one-shot	our structured prompt	–	–	–	–
FMISeg	supervised LMIS	original MedCLIP-SAMv2-style prompt	–	–	–	–
FMISeg	supervised LMIS	our structured prompt	–	–	–	–
LFAENet-TGFS	supervised LMIS	original MedCLIP-SAMv2-style prompt	85.41	76.92	81.90	74.26
LFAENet-TGFS	supervised LMIS	our structured prompt	86.51	78.11	83.86	76.15

The table supports the cross-dataset claim but should not be read in isolation. The following subsections isolate prompt format, prompt semantics, frequency handling, and CT-domain behavior. This separation is important because our contribution is not a new standalone visual backbone, but a language-conditioned extension of FAENet-style frequency refinement and a structured prompt protocol for supervised LMIS.

4.3. Prompt-Template Transfer on Brain MRI and Breast Ultrasound

The Brain MRI and Breast Ultrasound experiments require a separate prompt analysis because MedCLIP-SAMv2 is a promptable CLIP-SAM-style pipeline, whereas LFAENet-TGFS and FMISeg are supervised LMIS models trained with image-text pairs. Therefore, the fair question is not whether one prompt string is universally better, but whether a prompt format is compatible with a model family. Table 3 makes this explicit.

This experiment is intended to validate the prompt protocol rather than to claim that one prompt style is universally superior. The expected behavior is model-dependent: MedCLIP-SAMv2 may remain competitive with short CLIP-friendly prompts, while supervised LMIS models are expected to benefit from structured prompts that explicitly encode count, laterality, and coarse location. A positive result here would justify using our structured prompts for the Brain/Breast rows in Table 2.

4.4. Controlled QaTa-COV19 Ablations

QaTa-COV19 provides the cleanest controlled setting because each image is paired with a structured prompt describing laterality, lesion count, and location. Table 4 keeps only the rows needed to support the main mechanistic story: visual-only baselines, text-guided TGFS variants, prompt interventions, and text-encoder/fusion comparisons. Exhaustive run rankings are better placed in the supplementary material.

Three observations follow from Table 4. First, text-guided frequency decoding improves over visual-only FAENet baselines, indicating that the gain is not explained by the inherited frequency backbone alone. Second, native prompts consistently outperform empty, shuffled, and generic prompts, so the text branch is sensitive to prompt content and image-text correspondence. Third, encoder-decoder conditioning gives a modest but consistent gain in the scratch-family runs, while the ResNet50 decoder variant provides a strong architecture-aligned setting.

4.5. Frequency-Prior and Sub-band Diagnostics

Since the visual frequency machinery is inherited from FAENet, the ablation must focus on what TGFS changes: how language-conditioned decoding should handle individual Haar sub-bands. Table 5 combines two compact frequency diagnostics: HH-prior handling and one-band removal.

The HH-prior block shows that blindly keeping the diagonal high-frequency band is not optimal; hard suppression is strong, and learned retention performs best among completed rows. The drop-band block suggests that LL carries

Table 4

Controlled QaTa-COV19 ablations. Rows are grouped by experimental question rather than by run order. Best Dice/IoU within each group is in **bold**.

Group	Variant	Visual	Text	Fusion	Prompt	Dice	IoU
<i>A. Visual-only vs. text-guided frequency decoding</i>							
A	FAENet visual-only	scratch	–	–	–	77.57	67.65
A	FAENet visual-only	ResNet50	–	–	–	77.90	67.81
A	TGFS decoder	ResNet50	CXR-BERT	decoder	native	80.30	70.93
A	TGFS decoder	ResNet50	lightweight	decoder	native	82.24	73.43
<i>B. Prompt semantics</i>							
B	Native prompt	scratch	lightweight	both	native	82.12	73.05
B	Empty prompt	scratch	lightweight	both	empty	76.56	66.64
B	Shuffled prompt	scratch	lightweight	both	shuffle	75.85	65.75
B	Generic prompt	scratch	lightweight	both	generic	75.99	65.75
B	Native prompt	scratch	CXR-BERT	both	native	81.55	72.42
B	Empty prompt	scratch	CXR-BERT	both	empty	76.42	66.47
B	Shuffled prompt	scratch	CXR-BERT	both	shuffle	76.45	66.27
<i>C. Fusion locus and text encoder</i>							
C	Decoder-only	scratch	lightweight	decoder	native	81.63	72.43
C	Encoder–decoder	scratch	lightweight	both	native	82.12	73.05
C	Decoder-only	scratch	CXR-BERT	decoder	native	81.25	71.98
C	Encoder–decoder	scratch	CXR-BERT	both	native	81.55	72.42

Table 5

Frequency-prior and sub-band diagnostics on QaTa-COV19 using the ResNet50 TGFS decoder family. Best result within each block is in **bold**.

Block	Variant	Dice	IoU
<i>A. HH-prior handling</i>			
A	Keep HH	81.57	72.49
A	Zero HH	82.08	73.10
A	Learned HH retention	82.24	73.43
<i>B. Drop one wavelet band</i>			
B	Full LL/LH/HL/HH	81.57	72.49
B	w/o LL	81.08	71.85
B	w/o LH	81.63	72.59
B	w/o HL	82.10	73.24
B	w/o HH	81.96	72.85

important coarse lesion extent, while individual high-frequency bands are less uniformly beneficial and are better handled through soft or learned retention rather than a fixed rule.

4.6. Secondary MosMedData+ Diagnostic

MosMedData+ is used as a CT-domain diagnostic rather than the primary evidence for prompt semantics. Table 6 reports a compact subset of the M0–M8 matrix: the visual-only row, the decoder-only baseline, the encoder–decoder fusion row, the simple-text row, and the learnable-prior row. The full matrix is better suited for the supplementary material.

The MosMed diagnostic supports two conservative conclusions. First, the text branch and frozen CXR-BERT prior are important under CT domain shift. Second, fixed frequency assumptions are not optimal: the largest improvement in the matrix comes from learnable frequency-prior handling. Because empty and shuffled prompt behavior on

Table 6

Compact MosMedData+ diagnostic. The table keeps representative rows from the M0–M8 matrix and omits exhaustive prompt-policy runs from the main paper.

Case	Setting	Dice	IoU
M1	Visual-only FAENet, no text branch	42.18	31.27
M0	Decoder TGFS + frozen CXR-BERT + fixed frequency priors	65.42	51.29
M5	Encoder-decoder fusion + frozen CXR-BERT + fixed priors	66.95	52.89
M6	M5 with lightweight text encoder instead of CXR-BERT	57.16	43.35
M8	M5 with learned HH, learnable LF/HF scale, and learnable sharpening	70.08	57.34

MosMedData+ is less clean than on QaTa-COV19, we do not use MosMedData+ as the main semantic-prompt evidence.

4.7. Qualitative Cross-Dataset Analysis

Finally, qualitative examples should connect the quantitative tables to the model behavior. Figure 6 should include successful and failure cases from the four datasets. The goal is not to show every baseline, but to demonstrate where the structured prompt and frequency-aware decoder help: localized lesions, laterality-sensitive cases, blurry boundaries, and small foreground regions.

4.8. Summary of Experimental Findings

The experiments support a focused narrative. The main benchmark shows that LFAENet-TGFS is competitive across four datasets under a unified structured prompt protocol. The Brain/Breast prompt-template table isolates a second contribution: the structured QaTa/MosMed-style prompts are a valid interface for supervised LMIS and should not be conflated with MedCLIP-SAMv2-style prompts. The QaTa ablations provide the strongest semantic evidence: native prompts outperform empty, shuffled, and generic prompts. The frequency diagnostics show that TGFS benefits from learnable frequency-prior handling rather than hard rules over HH. The MosMed diagnostic complements this story by showing that CT transfer is sensitive to CXR-BERT and learnable frequency priors, while prompt semantics are cleaner on QaTa-COV19 than on MosMedData+.

5. Discussion

Main finding. The central hypothesis of this work is that a clinical prompt can directly select visual frequency components during mask reconstruction, and that this selection is useful rather than cosmetic. The strongest evidence comes from the controlled QaTa-COV19 ablations (Table 4): native prompts outperform empty, shuffled, and generic prompts by a clear margin under the same architecture and threshold protocol. Because only the text content changes across these rows, the improvement reflects the model using prompt *semantics*, not merely the presence of a text branch.

Text-guided frequency decoding adds more than a text channel. Relative to the visual-only FAENet baseline, adding the language-conditioned TGFS decoder improves segmentation on QaTa-COV19 (Table 4), and the gain is not explained by the ImageNet-pretrained encoder alone, since the visual-only row already uses the same encoder. Conditioning both the encoder and the decoder is consistently at least as strong as decoder-only conditioning, which suggests that language is useful both while early visual frequency evidence is extracted and while the mask is reconstructed. We read this as support for placing text conditioning inside the frequency-refinement pathway rather than as a late spatial-attention add-on.

Structured prompts transfer across modalities. On Brain MRI and Breast Ultrasound, the task-native structured prompt outperforms generic class-name and descriptive prompt templates under the same model (Table 3). This

indicates that a supervised language-guided model benefits from prompts aligned with the dataset and the mask definition, rather than from generic class phrases alone. The result should be read as a statement about prompt-format compatibility with a supervised model family, not as a universal claim that one prompt string is always better; a promptable foundation-model pipeline may behave differently under the same two prompt styles.

Frequency priors are better learned than fixed. The frequency diagnostics (Table 5) show that learned diagonal-band (HH) retention performs best, ahead of both hard suppression and naive keeping, while removing the low-frequency (LL) band degrades results the most. Together these observations argue against a fixed high-frequency prior: the coarse LL structure carries essential lesion extent, whereas the contribution of individual high-frequency bands is better decided by a learned, per-stage retention scale than by a single global rule.

CT transfer exposes a different bottleneck. MosMedData+ behaves differently from the chest-radiograph benchmark (Table 6): the largest improvement in the diagnostic comes from learnable frequency-prior handling, and prompt-semantic effects are less clean than on QaTa-COV19. Under CT domain shift, the frequency-prior design rather than prompt semantics appears to be the dominant lever. Since MosMedData+ is used here as a secondary robustness diagnostic, we treat this as a bound on generality rather than as central evidence for prompt semantics.

Broader insight. A recurring lesson across these experiments is that adding language to a segmentation model does not by itself guarantee semantic grounding. The guidance has to be verified through prompt interventions, and the frequency machinery has to be allowed to adapt to the modality. This distinction—between a text branch that is merely present and one whose content demonstrably moves the mask—is useful for language-guided medical segmentation more broadly.

Limitations. Several limitations remain. First, some entries in the comparison and prompt-intervention analyses are still to be completed under the unified protocol, and are marked “—” in the corresponding tables; the prompt-semantic evidence is therefore strongest on QaTa-COV19 and weaker on MosMedData+. Second, LFAENet-TGFS applies frequency selection at the intermediate feature level rather than to the raw input image, so its conclusions bear only indirectly on raw-image low-/high-frequency dual-branch designs. Third, the qualitative panels are provided as layout placeholders and should be replaced with final renders before publication. Finally, the structured prompt is a controlled text interface rather than a free-form radiology report, which bounds claims about behavior on unconstrained clinical text.

Future work. Future work should proceed in three directions. First, completing the remaining prompt-intervention and comparison runs under a single protocol would strengthen the semantic-grounding evidence, especially on the CT domain. Second, the frequency-prior handling could be extended from a learned HH scale to fully stage-wise, band-wise learnable priors, so that the model decides sub-band retention jointly with the prompt. Third, the structured prompt interface could be extended toward free-form reports and additional imaging modalities, testing whether text-guided frequency selection continues to help when the language is less templated.

6. Conclusion

We presented LFAENet-TGFS, a language-conditioned extension of FAENet-style frequency-aware segmentation. Unlike conventional language-guided methods that use text mainly as a semantic or spatial selector, LFAENet-TGFS injects clinical prompts into feature-level wavelet refinement: in each TGFS block, the pooled text vector modulates the Haar sub-bands $LL/LH/HL/HH$ while token-level embeddings provide a spatial grounding signal, so language controls both what frequency evidence is emphasized and where it is applied. A unified structured prompt format lets a single architecture serve four datasets across two imaging families.

With this single design, LFAENet-TGFS attains the best reported accuracy on Brain MRI, Breast Ultrasound, and MosMedData+ and the best IoU on QaTa-COV19, while controlled QaTa ablations isolate the contribution of text-guided decoding and the learned high-frequency prior. The current evidence is strongest on structured prompts; future work will complete the prompt-intervention and mechanism ablations and learn stage-wise frequency priors.

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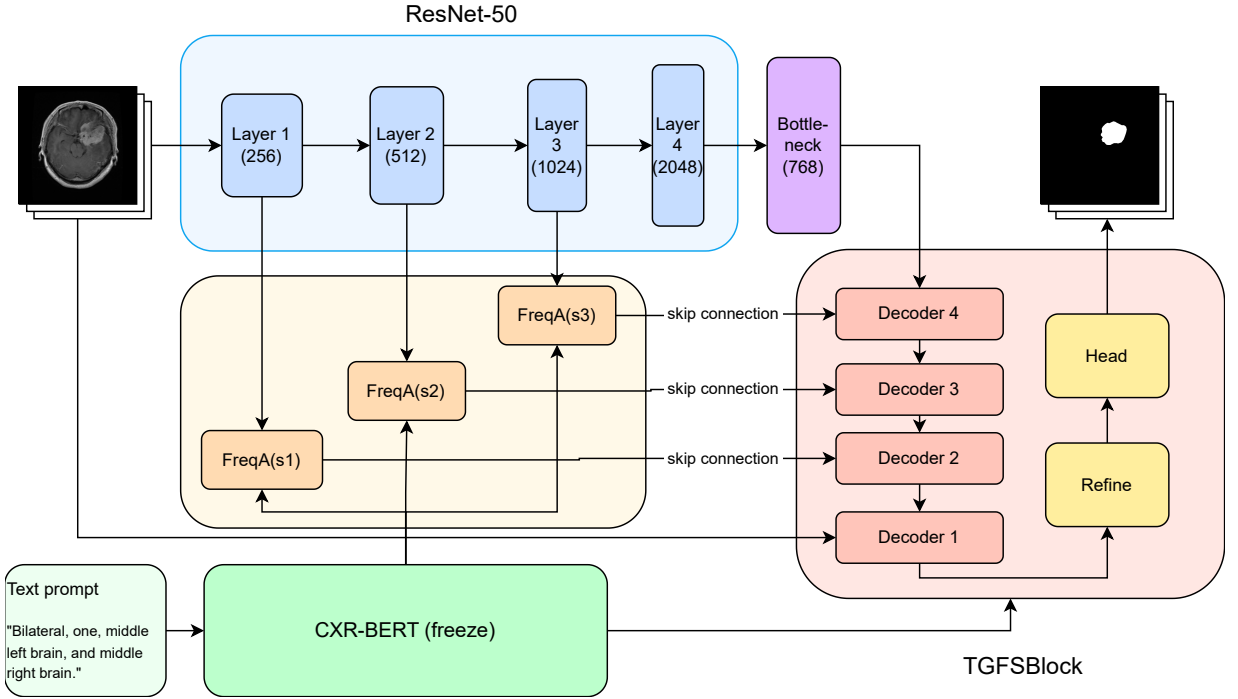


Figure 1: Overall architecture of LFAENet-TGFS. The visual stream follows a ResNet-50 encoder with FAENet-style feature-level Haar wavelet refinement. The text stream uses a frozen clinical language encoder to produce a pooled text vector and token embeddings. Text-conditioned TGFS blocks are applied in the encoder, bottleneck, and decoder. The model preserves the four sub-bands $LL/LH/HL/HH$ inside TGFS and treats the diagonal HH component with a learned retention scale.

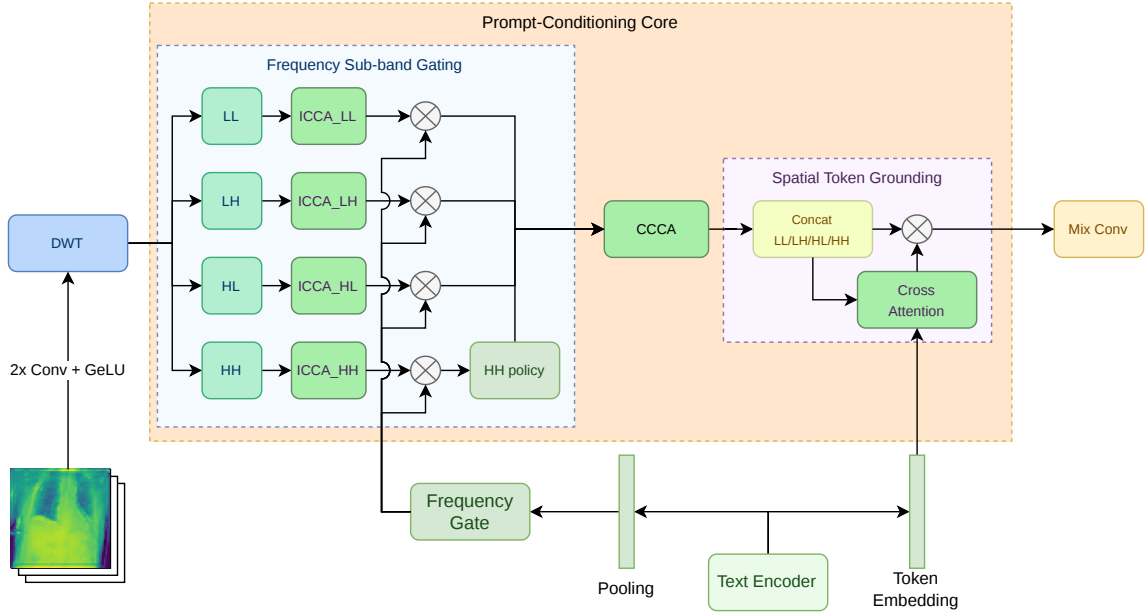


Figure 2: Prompt-conditioned frequency and spatial control. The pooled text vector generates global gates over the four wavelet sub-bands, while token embeddings interact with visual features to produce a spatial grounding mask M_s . Gates and spatial masks are applied across the encoder, bottleneck, and decoder.

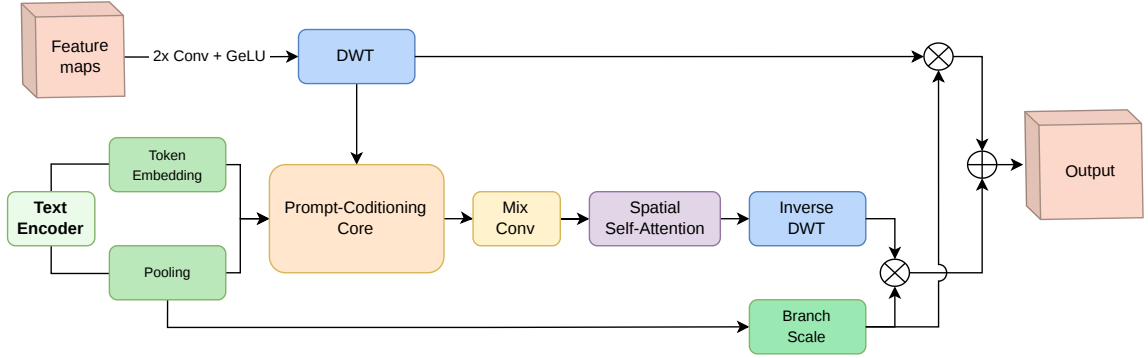


Figure 3: Text-Guided Frequency Selection block. TGFS decomposes an intermediate feature into $LL/LH/HL/HH$, applies per-band ICCA, uses the pooled text vector to generate sub-band gates $\alpha_{LL}, \alpha_{LH}, \alpha_{HL}, \alpha_{HH}$, performs CCCA across gated sub-bands, applies a token-grounded spatial mask M_s , and reconstructs the feature through inverse DWT. The diagonal HH component is retained with a learned scale.

Placeholder: Brain/Breast prompt-template qualitative figure.

Suggested rows: one Brain MRI case and one Breast US case. Suggested columns: input, ground truth, MedCLIP-SAMv2 with its original prompt, MedCLIP-SAMv2 with our structured prompt, LFAENet-TGFS with MedCLIP-style prompt, LFAENet-TGFS with our structured prompt, and error map.

Figure 4: Prompt-template diagnostic on Brain MRI and Breast Ultrasound. The figure should show whether the structured prompt better matches supervised LMIS models and whether MedCLIP-SAMv2 behaves differently under the two prompt styles.

Placeholder: QaTa prompt intervention.

Suggested columns: input image, prompt text, ground truth, native-prompt prediction, empty-prompt prediction, shuffled-prompt prediction, and error map. The image, checkpoint, and threshold should be fixed; only the text prompt is changed.

Figure 5: Prompt intervention on QaTa-COV19. The figure visually supports Table 4 by showing that changing only the prompt can alter the predicted mask and spatial error pattern.

Placeholder: Cross-dataset qualitative segmentation.

Suggested rows: QaTa-COV19, MosMedData+, Brain MRI, and Breast US. Suggested columns: input, structured prompt, ground truth, prediction, and error map. Add one extra row for a failure case if space allows.

Figure 6: Qualitative segmentation across four datasets. The figure should show how LFAENet-TGFS behaves under the unified structured prompt protocol and highlight both successful cases and typical failure modes.